

Gizem Toplu-Tutay

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Academic Appointment

Research Fellow (Postdoctoral), Georgetown University | 2026–present

The Earth Commons Institute, Washington, DC. Research mentor: Roshanak Nateghi.

Education

Ph.D., Operations Research and Industrial Engineering | 2026

The University of Texas at Austin. Dissertation: Multi-Objective Stochastic Optimization for Resilience Planning in Power and Healthcare Systems. Advisors: John J. Hasenbein and Erhan Kutanoglu.

M.S., Operations Research and Industrial Engineering | 2024

The University of Texas at Austin.

B.S., Civil Engineering | 2013

Boğaziçi University, Istanbul, Türkiye. Highest honors; ranked third. Thesis: Optimizing Wind Energy and Storage: A Financial and Operational Investment Analysis.

Research Interests

Methods: Stochastic and robust optimization; mixed-integer programming; multi-objective optimization; decomposition algorithms.

Applications: Infrastructure resilience; climate and disaster risk; power systems; societal impacts of disasters and service disruptions.

Publications

Published Journal Papers:

- **Toplu-Tutay, G.**, Hasenbein, J. J., and Kutanoglu, E., “[Impact of Power Outages Depends on Who Loses It: Equity-Informed Grid Resilience Planning via Stochastic Optimization](#),” *Socio-Economic Planning Sciences*, Vol. 95, 102036, pp. 1–16, 2024.
- Kim, K. Y., **Toplu-Tutay, G.**, Kutanoglu, E., and Hasenbein, J. J., “[A Stochastic Optimization Model for Patient Evacuation from Health Care Facilities during Hurricanes](#),” *International Journal of Disaster Risk Reduction*, Vol. 108, 104518, pp. 1–18, 2024.

Submitted Manuscripts and Preprints:

- **Toplu-Tutay, G.**, Kammer-Kerwick, M., Hasenbein, J. J., and Kutanoglu, E., “[Multi-Objective Planning for Healthcare Facility Resilience: Mitigate Now or Respond Later?](#),” submitted to *European Journal of Operational Research*, arXiv:2608.26304, 2026.
- **Toplu-Tutay, G.**, Hasenbein, J. J., and Kutanoglu, E., “[Social Vulnerability and Long-Run Tropical Cyclone Mortality: Evidence from Texas](#),” submitted to *Natural Hazards*, preprint, 2026.

Peer-Reviewed Conference Papers:

- Das, J., **Toplu-Tutay, G.**, Gulen, G., and Young, M., “[Power Grid Dispatch Modeling for the Prospective](#)

Current Trends Scenario for the ERCOT Service Area of Texas,” *IEEE Texas Power and Energy Conference (TPEC)*, pp. 1–6, 2025.

- **Toplu-Tutay, G.**, Hasenbein, J. J., and Kutanoglu, E., “Scenario-based Optimization Model for Long-term Healthcare Infrastructure Resilience against Flooding,” *IIEE Annual Conference Proceedings*, pp. 1–6, 2022.

Work in Progress:

- *The Cost of Planning for the Wrong Climate Future: Climate-Robust Resilience Investment under Deep Uncertainty* (with John J. Hasenbein, Erhan Kutanoglu, and Roshanak Nateghi).
- *Risk Analysis in the Energy Transition* (with Roshanak Nateghi).

Research Experience

Doctoral Research Assistant | The University of Texas at Austin | 2021–2026

Optimization under uncertainty, infrastructure resilience, and disaster impacts.

- Developed stochastic and multi-objective optimization models for infrastructure resilience under extreme-event uncertainty, with applications to power-grid hardening, healthcare-facility protection, patient evacuation, and emergency resource allocation.
- Developed scalable solution approaches, including multi-cut Benders decomposition and Lagrangian methods, for large-scale resilience planning problems involving thousands of facilities and multiple disaster scenarios.
- Integrated infrastructure, hazard, climate, and social-vulnerability data to evaluate both system-level performance and the heterogeneous consequences of service disruptions across communities.
- Developed statistical and panel-data models to quantify long-run disaster impacts and examine how tropical-cyclone mortality varies with social vulnerability.
- Developed mixed-integer optimization models for electricity capacity planning and system operations under alternative generation portfolios, renewable-energy penetration, and energy-transition scenarios.
- Coupled optimization models with physics-based flood and storm-surge models to translate uncertain hazard conditions into infrastructure decision requirements.

Ph.D. Research Intern | National Renewable Energy Laboratory | May–August 2024

Golden, CO. Electric vehicle managed charging and electricity system planning.

- Contributed to incorporating electric vehicle charging flexibility into a national-scale electricity capacity planning model using detailed charging load profiles; developed data-processing and model-output analysis scripts.
- Used high-performance computing and collaborative Git/GitHub workflows for model development and analysis.

Teaching Experience

Teaching Assistant, Analytics for Supply Chain | Fall 2025

The University of Texas at Austin, McCombs School of Business. Supported course assessment and grading.

Teaching Assistant, Engineering Statistics | Fall 2022

The University of Texas at Austin, Department of Mechanical Engineering. Designed and led recitations; held office hours; graded assignments and exams. Responsible for a 25-student section.

Teaching Assistant, Learning Structures and Time Series | Spring 2021

The University of Texas at Austin, McCombs School of Business. Held office hours and graded assignments and exams for a graduate course in unsupervised learning and forecasting. Enrollment: 105.

Honors, Awards, and Research Support

- **University Graduate Continuing Fellowship**, The University of Texas at Austin, 2024–2025. Recipient of a one-year \$52,000 university fellowship for outstanding academic achievement.
- **Graduate Research Assistantship, Bureau of Business Research / IC² Institute**, The University of Texas at Austin, 2025–2026. Individually selected for research support.
- **IC² Institute Seed Grant, *Towards a Well-Being Economy: Advancing Equity in Social, Economic, and Health Systems***, 2022–2023. Contributed to proposal development and conducted the funded research under the faculty PI.
- **Comparing Electricity Options Project, Bureau of Economic Geology**, 2023–2024. Individually selected for research support.
- **NSF Award #1940308, *CoPe EAGER: Addressing Human-Centric Decision-Making Challenges from Coastal Hazards via Integrated Geosciences Modeling and Stochastic Optimization***, 2021–2022. Conducted the funded research under the faculty PI.
- **Highest Honors**, B.S. in Civil Engineering, Boğaziçi University, 2013.
- **Academic Excellence Scholarship**, 2008–2013.

SELECTED PRESENTATIONS

- **Invited seminars on optimization models for resilient infrastructure and emergency response under uncertainty**
 - INFORMS Student Chapter, Texas State University, June 2026.
 - Department of Engineering Technology & Industrial Distribution, Texas A&M University, January 2026.
 - Operations Research and Industrial Engineering, The University of Texas at Austin, April 2025.
 - Department of Industrial Engineering, Clemson University, January 2025.
- ***Equity-Informed Power Grid Resilience Planning via Stochastic Optimization***
 - Society for Risk Analysis Annual Meeting, Austin, December 2024.
 - INFORMS Annual Meeting, Seattle, October 2024 (Poster).
- ***Disparate Impacts of Extreme Flooding and Power Grid Failures in Different Communities***
 - INFORMS Annual Meeting, Indianapolis, October 2022.

- *A Scenario-based Optimization Model for Long-term Healthcare Infrastructure Resilience against Flooding*
 - IISE Annual Conference, Seattle, May 2022.
- *Electricity Market Developments*
 - Annual seminars at CLK Energy Investment, 2015–2017.

Professional Service and Memberships

Journal reviewing: Applied Energy; IEEE Transactions on Energy Markets, Policy, and Regulation; Computers & Industrial Engineering; Utilities Policy

Professional memberships: Society for Risk Analysis; INFORMS; International Association for Energy Economics (IAEE).

Conference organization: Organizer and committee member for IAEE international conferences and symposia, including Eurasian Conferences and the Southeast Europe Symposium, 2016–2019.

Summer school organization: IAEE Summer Schools on Electricity Markets and Regulation, Istanbul, 2015–2016.

University service: Chair, Tenant Advisory Board, University Housing and Dining, The University of Texas at Austin, 2023–2026.

Industry Experience

Founder | GISEMAI | Ankara, Türkiye | 2018–2020

Managed procurement, inventory, warehousing, and fulfillment; used demand and supplier lead-time analysis to support replenishment decisions.

Senior Strategic Planning Analyst | CLK Energy Investment | Istanbul, Türkiye | 2015–2017

Developed R-based electricity demand forecasts and scenario analyses for day-ahead, intraday, and balancing markets. Supported procurement, bidding, and risk analysis for government auctions and bilateral contracts.

Key Account Manager | ENERJISA | Istanbul, Türkiye | 2013–2014

Managed large electricity customers, analyzed market conditions, and negotiated contracts in coordination with technical teams.